

Personal Space Evaluation and Protection in Social VR

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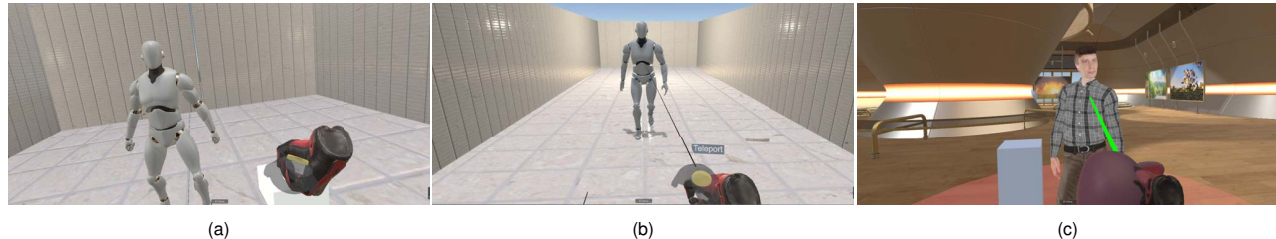


Figure 1: Scenes used in the experiment. (a) Standing test. (b) Approaching test. (c) Personal space protection test.

ABSTRACT

Social VR has been widely promoted and popularized in recent years. Due to the immersion characteristic of VR, although people do not have physical contact in the virtual space, they still have judgements on distance and may feel annoyed when their personal space is intruded. Social VR users want to communicate with their virtual friends in the space. Meanwhile, keep a comfortable distance from other avatars and protect their personal space. In this paper, we evaluate user's perception and comfortable level of personal distance and compare four different methods to protect personal space in social VR.

Index Terms: Human-centered computing—Interaction paradigms—Virtual reality—;

1 INTRODUCTION

Social VR allows several users in separated physical spaces to communicate and collaborate in the virtual environment. Public social VR systems (e.g. VRchat, Facebook social VR and Altspace VR) load multiple remote users into a shared virtual space for socializing and activities. Users can design their own virtual avatars by selecting styles to ensure that they can express their unique personalities.

Research on social VR mainly focuses on the influence of the social existence and coexistence of virtual avatars on the experience and participation. Results show that the higher the integrity of the social existence of virtual avatars, the better the user experience [2]. However, users may feel discomfort and arousal if the area around them (personal space) is intruded by other users. Factors such as gender, height etc. of avatars may influence user experience and user's personal space [3]. Hecht *et al.* study the shape of personal space in physical and virtual space and investigate that height and gender of the avatar have a great influence on the personal space, while approaching type does not matter [1]. However, the study of personal space protection in virtual environment is still lacking. What factors influence user's comfortable distance and how to protect personal space are vital in social VR design. In this paper, we evaluate influence of avatar's height on user's comfortable distance

in virtual space, and compare different ways to protect personal space in social VR.

2 EXPERIMENT DESIGN

In the experiment, we first evaluate the influence of avatar's height on user comfortable distance. Then propose schemes to protect personal space based on the evaluation results.

2.1 User comfortable distance

In order to study whether the avatar's motion has an influence on comfortable distance, we test user comfortable distance in standing and approaching state. We use a simulated mannequin with controlled height and state as the friend avatar to avoid influences of other factors. Screenshots of the experiment are shown in Figure 1 (a) and (b).

Standing test evaluates the influence of a static avatar's height on users' comfortable distance. The participant needs to stand in the yellow test area in the virtual space and learn how to trigger the three test buttons of "Accept", "Don't Accept" and "Next", as well as the operation mode of the distance adjustment slider. When a virtual agent appears, the participant needs to judge whether its height is acceptable by clicking the "Accept" or "Don't Accept" button. If the height is acceptable, a virtual slide bar will appear in front of him. The participant can adjust the distance between the agent and himself by sliding the bar. When the participant feels the distance is comfortable, click the "Next" button to complete the test. There are 10 virtual agents with different heights ranging between 1.08m-2.7m in the whole test, which appear one by one in a random order.

Approaching test evaluates the influence of an avatar's height and speed on users' comfortable distance. The participant needs to stand on the centerline of the corridor facing to the front and learn how to use rays and trigger keys to trigger a stop. An agent appears in front of the participant 15-meter away and approaches the participant at controlled speeds. When the participant feels uncomfortable with the approaching agent, the participant needs to point the rays at the virtual social objects and pull the trigger to trigger a stop event. There are 10 virtual agents with different heights ranging between 1.08m-2.7m in the whole test, which appears one by one in a random order.

2.2 Personal space protection

If an avatar is too close to the user, the user may feel his personal space intruded and uncomfortable. Therefore, when an avatar is approaching the user, it needs to stop at a threshold distance to protect personal space. However, the procedure to stop the avatar has

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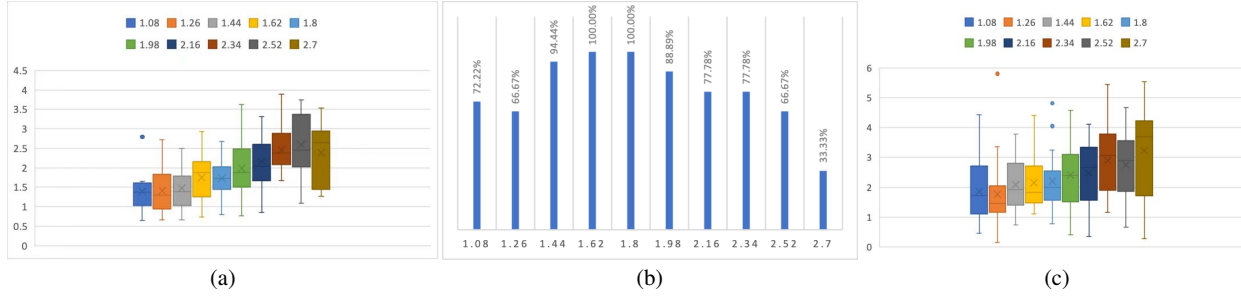


Figure 2: Results of the User comfortable distance test. (a) Comfortable distance of each agent height in the standing test. (b) User acceptance rate on agent height. (c) Comfortable distance of each agent height in the approaching test.

a great influence on user experience. In this test, we apply the results of the user comfortable distance test to designing rules for social VR applications to test the effectiveness of different methods on personal space protection. We show a screenshot of the experiment in Figure 1(c).

Before the test, the participant needs to practice grasping objects and starting events with vive controllers. When the test starts, there will be a table in front of the participant, on which some props are placed. In the distance, there will be four virtual agents at an equal distance. The participant needs to grasp a prop, select the target agent put the props on the table next to the agent, and then click to return to take the next prop.

In previous tests, we evaluate the comfortable distance of users in social VR. In this test, we design four different stopping mode to keep the distance between the agent and the user (UAD). In the first mode (slower), when UAD is less than the user's approaching comfortable, the user's moving speed will gradually decrease. In the second mode (forced to stop), when UAD is less than the user's standing comfortable distance, ray casting movement is forbidden. In the third mode (visual cue), when UAD is less than the user's standing comfortable distance, ray casting movement is forbidden and a red circle appears to remind the user that he may intrude on other user's personal space. In the fourth mode (pause), when UAD reaches the comfortable distance, the user's function of ray casting movement will be temporarily forbidden and recovers after 2 seconds. After the test, the participant is asked to choose one favorite mode and one least favorite mode.

2.3 Experimental Settings

We use an HTC Vive Head Mounted Display (2160 × 1200 screen resolution, 90Hz refresh rate, 110° horizontal FoV) to provide VR experiences for participants. We also provide participants with two Vive controllers to interact with the scene.

We recruited two groups of volunteers from college students and all participants are aged between 18 and 24 years old. In the user comfortable distance test, there were 18 participants (8 males and 10 females), among which 7 have VR-related experience and 11 have none VR-related experience. There are 16 participants in the personal space protection test (8 males and 8 females), among which 12 have VR-related experience and 4 have none VR-related experience.

3 RESULTS

Figure 2(a) shows user's comfortable distance ranging with different agent heights in the standing case. When the agent height is between 1.08m-2.52m, the comfortable distance increases with agent height, and when the agent height is 2.7m, the comfortable distance decreases slightly. ANOVA analysis (significance level of 5%) shows that agent height has a very significant correlation with the comfortable distance, with $F(22.24, 52.66) = 6.15$, and $p < 0.01$. As shown in Figure 2(b), agent height between 1.44m and 1.98m is generally

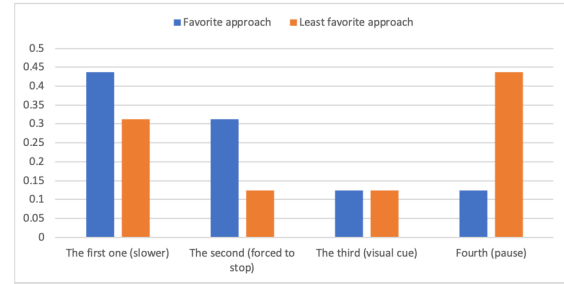


Figure 3: User response on favorite/least favorite approaching mode.

accepted by people (more than 85% of participants accept it), and acceptance rates at 1.62m and 1.80m are both 100%. We conclude that users may prefer avatars with a normal human height in VR. When the height of the agent is 2.7m, the acceptance decreases sharply, and only 33.33% of participants accept it. Figure 2(c) shows the results of comfortable distance ranging with different agent heights in the approaching case. Generally the comfortable distance increases with the increasing height of agent. We use the significance level of 5% to do one-way ANOVA analysis and find that agent's height has significant influence on comfortable distance, $F(35.39, 213.46) = 3.13$, $p = 0.0016 < 0.05$. We also find that the comfortable distance in the approaching test is larger than that in the standing test on average. We conclude that the agent moving state has a significant impact on the comfortable distance ($p = 0.006 < 0.01$).

Figure 3 shows the proportion of participants' choice of the most favorite and least favorite approaching mode. Although the number of participants who choose the first mode as their favorite is the largest, a lot of participants choose it as the least favorite. After short reviews with these participants, we find that participants have different perceptions of speed in a virtual environment, and these participants feel that they move too fast in the first mode.

4 CONCLUSION

In this paper, we evaluate the influence of avatar height on user's comfort level of distance and compare different ways to protect personal space in social VR. We hope this work can promote design guidelines of Social VR applications considering personal space protection. Future work will focus on studying more factors influencing comfortable distance and multi-user study in social VR.

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